S1 Text. Supplementary methods and reproducibility guide

Experimental context limits transfer of microprotein sequence scores. 10 November 2026.

This companion reflects the current preprint. The complete experimental-source, independent-transfer and native-evidence methods are in the main article. All analyses reuse published measurements; no new experiment or model fit was added during preprint preparation. The historical 31 October supplement is preserved for provenance and is superseded by this guide. Full Guo source methods have since been recovered and reviewed; raw reads and spectra have not been reprocessed.

1 Benchmark specifications

1.1 Dependence audit and score provenance

We grouped records linked by identical peptide sequences, a shared parental construct, available gene identities or overlapping catalogue entries. Engineered sequences shared by different parents connected those parents. We exhaustively aligned eligible pairs among native assay peptides, Yang natives, annotation-reference sequences and available ShortStop positive-reference counterparts. Local Smith–Waterman alignment used BLOSUM62, gap-open 10 and gap-extension 1. Primary edges required at least 60% identity over alignment columns and at least 80% coverage of **both** full sequences; a 40% identity graph was a declared sensitivity. Length and composition bounds only removed pairs that could not meet these criteria. Connected components defined split groups. These groups keep known linked records together during validation. They do not rule out distant homology or earlier exposure during pretraining.

The annotation-trained AAC classifier retained the original fixed Extra Trees configuration: 200 trees, maximum depth 10, minimum leaf 10, square-root feature sampling, balanced classes and seed 100. Whole reference pairs and linked reference split groups were removed to a fixed point if connected to any Guo or Yang parent. Seeds 101 and 102 were sensitivity analyses. Higher scores indicate canonical resemblance, opposite to the nORF Score convention [9]; this classifier was not trained on reporter measurements.

ShortStop v1.2.0 (28 October 2026) precomputed scores were joined only after independent GTF translation against hg38 verified exact peptide identity. The primary continuous score was mean SAM probability across released models. Published standard and stringent consensus calls were retained as distinct threshold analyses. Actual supervised-positive identifiers were distinguished from ToBePredicted records, which the inspected training code excludes from supervision. The 1,571 available positive-reference sequences are from the current reviewed release, which does not provide a complete archive of historical training sequences. Thus absence of exact matches does not establish complete training or pretraining separation.

The frozen PAP human30 model was reimplemented from safely parsed published numeric coefficients [14]. Primary transfer uses the final $\min(30, L)$ residues. Composition terms scale residue counts to 30 positions; terminal, pair and triplet terms follow the released positional definition. The full-peptide version uses normalized composition and is evaluated as a secondary analysis; PAP was not refitted. Deposited Guo classification/regression predictions were evaluated only on the additional natural sequences and their strict subset, not against other methods' held-out predictions on Guo's own training rows. Supplied PepScore values were used only for

the exact Yang constructs; unavailable transcript, domain and evolutionary inputs were left missing.

1.2 Endpoint models, order controls and nested validation

Models used five outer and three inner group folds. Groups were randomly tie-broken with seed 20261031, sorted by descending size and greedily assigned to the smallest fold without using outcomes. Ridge penalties $\{0.1, 1, 10, 100, 1000\}$ were selected by inner mean absolute error (MAE). Standardization and intercept fitting used each training split only. All representations shared the same outer folds. Predictions and fit receipts were saved for every held-out row.

The length model L contains log length and its square. A adds all 20 AAC frequencies. T adds first-five and last-five residue one-hot encodings, whole/N10/C10 hydropathy and PAP composition/terminal components. O adds 40 short-range property-product averages: eight default Moreau–Broto AAindex properties at lags 1–5, standardized across amino acids, then centered and scaled against 199 within-peptide shuffles. Shuffles preserve exact composition and the first and last five residues. Zero null variance produces a zero residual. This provides features for short cargos without imputing lag-30 values.

Three independent shuffled-order representations add the same 40 standardized features to T using additional shuffled realizations. They test whether residue order adds predictive information after composition and terminal residues are held fixed; the shuffled peptides have no measured outcomes and are not labelled experimental negatives. Raw order features, two extra fold seeds, 40% identity grouping and fixed Extra Trees regression for A/T/O were declared sensitivities. The tree regressors used 200 trees, depth 10, leaf minimum 10 and feature fraction 0.7. No outer-fold winner was substituted for the primary Ridge analysis.

Primary contrasts are O-minus-T rank correlation and T-minus-O MAE, with 1,000 operational-group bootstrap resamples conditional on saved predictions. These intervals keep the fitted models fixed and therefore omit uncertainty from retraining. Additional intervals and subgroup results are descriptive; no secondary significance claims or multiplicity-adjusted discovery list is made. No equivalence test was performed. We report the median absolute difference between biological replicates and half that value to show the scale of replicate variation. Neither is a formal noise floor or a threshold for biological relevance.

1.3 Calibration, perturbation prediction and independent transfer

Frozen scores were mapped to reporter scale using isotonic regression fitted to training WTs within each group fold. Direction was determined by training correlation and recorded, including reversed AAC-resemblance maps. Raw score MAE was not compared across incompatible scales. For mutations, each parent's WT and variant were predicted with the **same parent-held-out** model or isotonic map trained on mutagenesis-library WTs. We report raw rank differences as well as calibrated prediction changes. Zero change and an out-of-group training-family mean are error baselines. Replicate-specific outcome changes were analyzed separately.

Secondary mutation-trained Ridge models use mutant-minus-WT changes in A/T/O features and grouped nested evaluation within each family. Opposite-family transfer trains on the other edit family after excluding all held-out parent groups. These tests distinguish using an unmodified-peptide score to predict an edit, training directly on that type of edit, and applying a model trained on a different edit type.

Before scoring Yang outcomes, main predictions and model choices were saved in a validation-release receipt. Primary direct transfer trained on MP–mCherry WTs, whose cargo has a free N terminus and masked C terminus, as an approximate context choice for a C-terminal FLAG construct. Actual FLAG-fusion sequence features were primary; native-only features and GFP–MP training were declared sensitivities. Every training group connected to any Yang construct was removed from these direct-model fits. We did not fit, invert, calibrate or select models on the Yang outcomes. The input representation remains an approximation because neither Guo reporter construct is identical to a FLAG construct. Analyses report all 37, the 27 without exact Guo overlap, and the 20 passing strict separation from all retrieved Guo natural cohorts. Rank correlation and AUROC at the authors' 5% expression threshold are reported; absolute MAE is not reported across assays with different measurement scales.

2 Guide to supporting files

S1–S18 Data are logical sections of one curated public reproducibility bundle, rather than eighteen separately supplied original-source archives. The repository is `https://github.com/Kohze/microprotein-score-transfer`. The frozen data and analysis release is v0.1.1 (commit `1a22e80f60b46fefa8f370100b2e322457f3f518`). This appendix is supplied at `paper/S1_Text.pdf`, with editable source at `paper/S1_Text.tex`; both are linked from the repository README. `SUPPORTING_DATA_MAP.csv`, supplied with this supplement and at the repository root, gives exact paths for every label. Paths below are relative to the bundle root. Directory entries include their frozen tables, protocols and original analysis scripts. Upstream-dependent scripts are historical source-reconstruction records; their presence does not certify a clean-machine replay. Original papers, figures, workbooks, third-party implementations and model weights must be obtained from the cited providers under their own terms.

S1 Data. Reporter prediction metrics.
`analysis/stability_benchmark_2026-10-31/results/metrics.csv`;
`analysis/stability_benchmark_2026-10-31/results/common_coverage_metrics.csv`;
`analysis/stability_benchmark_2026-10-31/results/conditional_intervals.csv`.

S2 Data. Paired-edit summaries.
`analysis/stability_benchmark_2026-10-31/results/metrics.csv`;
`analysis/stability_benchmark_2026-10-31/results/native_contrasts.json`.

S3 Data. Selected expression-panel predictions and coverage.
`analysis/stability_benchmark_2026-10-31/results/Yang_validation.parquet`;
`analysis/stability_benchmark_2026-10-31/validation_assay_records.parquet`;
`analysis/stability_benchmark_2026-10-31/results/Yang_full_ShortStop_coverage_audit.csv`.

S4 Data. Reporter row-level predictions.
`analysis/stability_benchmark_2026-10-31/results/native_GFP_MP.parquet`;
`analysis/stability_benchmark_2026-10-31/results/native_MP_mCherry.parquet`;
`analysis/stability_benchmark_2026-10-31/results/native_GFP_MP_mutagenesis.`

parquet;
analysis/stability_benchmark_2026-10-31/results/additional_transfer.parquet.

S5 Data. Verified mutation pairs and predictions.
analysis/stability_benchmark_2026-10-31/results/mutation_predictions.parquet.

S6 Data. Exploratory prediction diagnostics. analysis/stability_benchmark_2026-10-31/results/exploratory_top_decile_prioritization.csv; analysis/stability_benchmark_2026-10-31/results/mutation_coupling_diagnostics.csv; analysis/stability_benchmark_2026-10-31/results/exploratory_additional_substitution_metrics.csv.

S7 Data. Protocols, dependence records and verification.
analysis/stability_benchmark_2026-10-31/INPUT_FREEZE.json;
analysis/stability_benchmark_2026-10-31/EXECUTION_PROTOCOL.txt;
analysis/stability_benchmark_2026-10-31/sequence_dependence_groups.parquet;
analysis/stability_benchmark_2026-10-31/results/fit_receipts.json;
verify_saved_results.py; provenance/UPSTREAM_SOURCE_REGISTRY.json.

S8 Data. Same-peptide reporter transfer. analysis/paired_context_2026-11-01/.

S9 Data. Canonical-tile transfer.
analysis/novelty_audit_2026-11-01/independent_test/.

S10 Data. Independent paired effects and published baselines.
analysis/novelty_audit_2026-11-01/independent_interventions/;
analysis/novelty_audit_2026-11-01/published_predictor_comparison/;
analysis/novelty_audit_2026-11-01/pap_neural_comparison/.

S11 Data. Terminal exposure and matched-input controls.
analysis/novelty_audit_2026-11-01/terminal_predictor_comparison/.

S12 Data. Source edit filtering and exact candidate links.
analysis/novelty_audit_2026-11-01/native_function_bridge/.

S13 Data. Native-candidate evidence and RNA-context annotations.
supporting/S13/.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

S14 Data. Current MS evidence annotations. supporting/S14/.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

S15 Data. Brain-context sequence annotations. `supporting/S15/`.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

S16 Data. Cardiac and cancer-study candidate annotations. `supporting/S16/`.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

S17 Data. Regulatory, screen and structural-control annotations. `supporting/S17/`.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

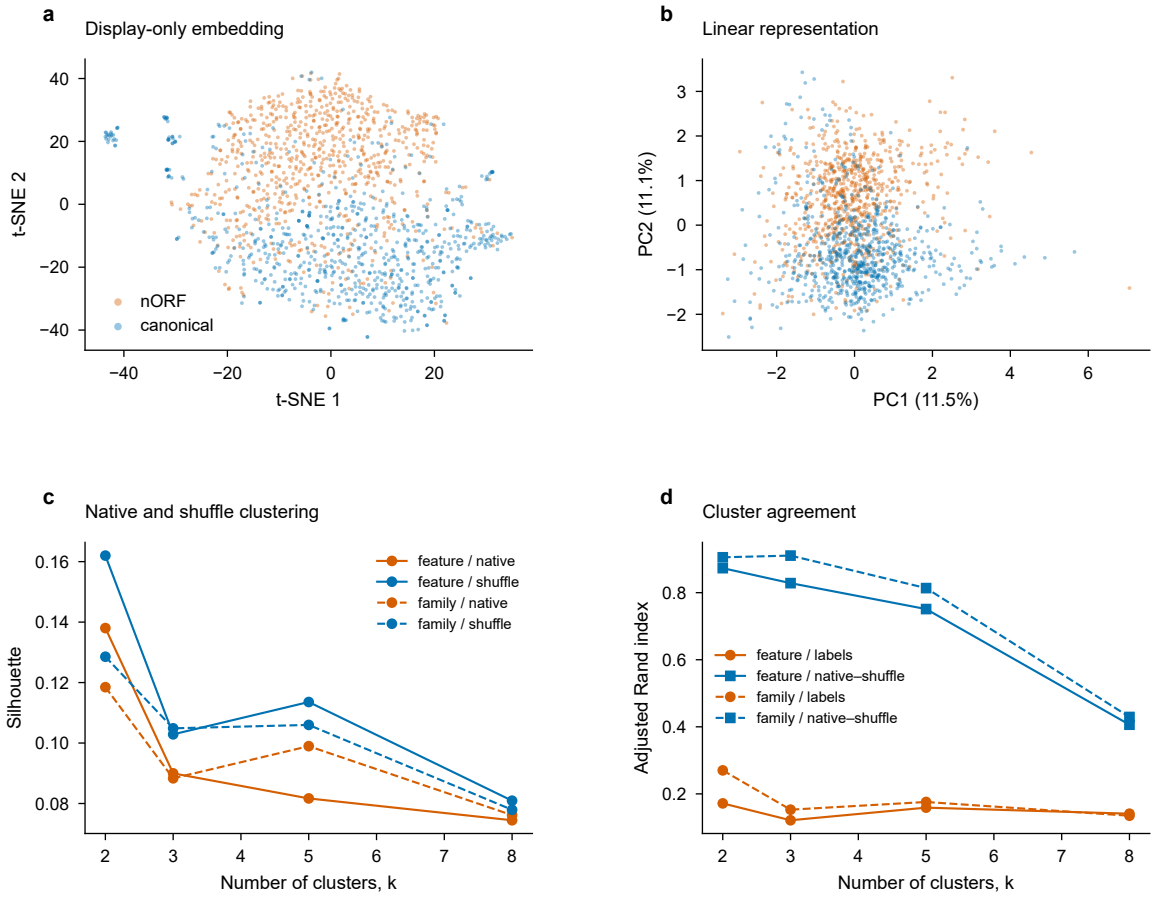
S18 Data. Primary validation precedent annotations. `supporting/S18/`.

Derived candidate annotations and recorded source-join checks are included. Original source documents and experiments are not replayed by the root verifier.

3 Reproduction and source availability

The supported offline entry point is `verify_saved_results.py` at the bundle root, with dependencies in `requirements-verification.txt`. It checks every manifest member and recomputes saved point metrics without fitting models or contacting the network. It does not recompute bootstrap intervals, recover original workbooks, run pretrained models or independently validate native biology. `REPRODUCING.md` describes the historical source workflows; `provenance/UPSTREAM_SOURCE_REGISTRY.json` identifies upstream inputs. The added S13–S18 derived annotations are hash-checked, but their original source joins are not independently rerun by this offline verifier.

S1 Text contains the detailed methods and results. S1 Fig shows controlled catalogue clustering; S2 Fig covers assay availability and dependence; S3 Fig shows order-feature increments; S4 Fig shows within-study paired edits; S5 Fig presents the selected FLAG-expression challenge; and S6 Fig shows calibration and conditional-order controls. The corresponding artwork and frozen plotting tables are in `paper/figures/`.



S1 Fig. Controlled catalogue clustering context. Controlled catalogue clustering of length-matched canonical and noncanonical proteins. a,b: illustrative t-SNE/PCA of the same 1,488 matched proteins. c,d: native/shuffle silhouettes and agreement across k and weighting. The fixed computational rerun reproduced these controlled maps/memberships; the original cluster memberships could not be recovered, and functional labels were not tested. UMAP was not rerun.

4 Detailed benchmark, transfer and exploratory results

4.1 Annotation resemblance and reporter accumulation are different targets

Training on measured reporter outcomes improved prediction substantially. We first define the eligible evaluation cohorts, then compare annotation resemblance, outcome-trained models and the additional information supplied by residue order.

4.1.1 Dependence controls materially reduce transfer coverage

The similarity audit covered 4,590 unique native/reference peptides and 2,370,772 eligible alignments. It identified 1,708 primary and 5,319 sensitivity similarity edges. Across all source types, primary components numbered 3,474, with a largest component of 254. The 1,072 development constructs occupy 800 groups; the largest contains 156 constructs. Large linked groups are retained, not split to improve precision (Fig S2).

Excluding assay-connected components and whole matched reference pairs left 491 balanced annotation pairs from the original 744. No retained pair connects to a retrieved assay component. Of the 801 additional natural sequences, 602 remain separated from the development groups. Ten Yang peptides exactly match retrieved Guo natural sequences, and related-sequence/gene links reduce its strict subset further, from 27 to 20. Current ShortStop-positive counterparts have no exact matches to the 1,072 development or 801 additional peptides, but historical and remote-similarity limitations remain.

4.1.2 Annotation resemblance and reporter ranking diverge

The independently trained AAC resemblance score correlated negatively with the native GFP–MP outcome ($\rho = -0.292$, 95% interval -0.344 to -0.237). The three fixed classifier seeds gave $\rho = -0.292$ to -0.293, so the negative association did not depend on the chosen seed. ShortStop's mean SAM probability had little monotonic association ($\rho = -0.018$, interval -0.077 to 0.041; $n = 1,057$). Frozen PAP C30 reached $\rho = 0.681$ and its full-sequence secondary transfer 0.761 (Fig 1).

Threshold prioritization gives a different result from continuous ranking. Standard SAM calls ($n = 76$) had median reporter score 0.593 versus 0.436 in other scored cargos, but were longer (median 57 versus 38 aa). A descriptive rank association after residualizing both continuous score and outcome on log length and its square was 0.044. There were only 23 consensus-90 and six consensus-100 calls; their summaries are retained without a strong inference. These data do not establish that SAM prioritization is useless, or that PRISM-labelled products lack function. They show that the continuous score is not a monotonic proxy for this reporter endpoint.

Direct training on the experimental target changed performance substantially (Table 1). Length alone achieved $\rho = 0.207$, AAC model A 0.747, and terminal-aware T 0.780. The strict additional subset preserved useful composition prediction ($\rho = 0.719$). On the same 602 sequences, ShortStop had $\rho = -0.085$ among its 595 covered peptides, AAC resemblance -0.292, PAP C30 0.630, and the released Guo regression 0.737. The Guo score is a useful deposited comparator, but released-training disjointness does not prove blinded external evaluation. No Guo in-sample training estimate is used in Table 1. The common-coverage analysis gives the same pattern: on 595 strict additional cargos, A, T, O and Guo regression

achieve $\rho = 0.720$, 0.713, 0.714 and 0.737, respectively, versus -0.292 for AAC resemblance and -0.085 for ShortStop. The common native comparison contains 1,057 cargos.

4.1.3 Adding residue-order features to T gives small, inconsistent improvements

The primary O-minus-T rank gain was 0.0040 (95% interval -0.0046 to 0.0115), with MAE reduction 0.00077 (interval -0.00149 to 0.00332). Native order outperformed the three shuffled representations in the primary point estimates, but its gain over T changed with evaluation context (Fig S3). Two additional split seeds gave rank gains of -0.00015 and 0.00517; 40% identity grouping gave 0.00594. In the opposite orientation O performed worse than T ($\Delta\rho = -0.0102$), as it did for mutagenesis-library WTs (-0.0076). Fixed tree-regression sensitivity also showed no O advantage over T ($\rho = 0.813$ versus 0.814). Raw order features gave $\rho = 0.785$ in the primary task.

The median absolute native replicate disagreement was 0.0545; half this is 0.0273, much larger than the primary point MAE reduction, but it is not a formal minimum useful effect or an equivalence margin. The intervals and sensitivity analyses show that adding these order features gives small and inconsistent improvements. They do not prove that residue order is biologically irrelevant. All 261 cargos at most 30 aa were retained: their T/O correlations were 0.819/0.817, versus 0.761/0.767 for 811 longer cargos. Complete-replicate, complexity and ORF-context summaries are supplied without choosing a winning subgroup. Only ten development peptides met the low-complexity flag, too few to support a separate conclusion.

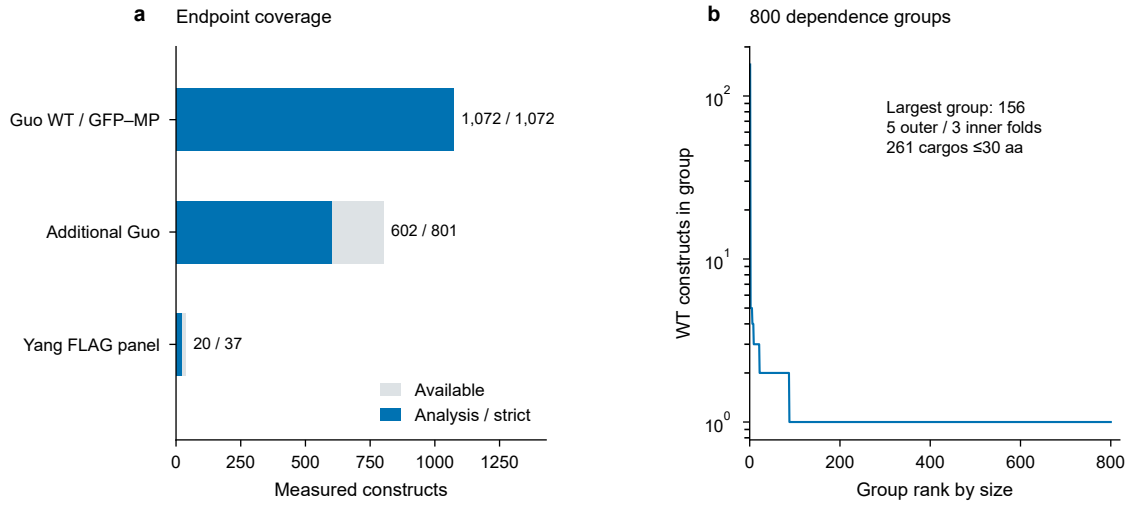
4.1.4 The catalogues remain distinguishable after matching, without establishing functional classes

Only 9,369 of 27,670 updated nORFs (33.9%) meet the 31-residue domain of the default all-family panel, compared with 19,534 of 19,563 canonical representatives. After exact length matching, AAC discrimination reached AUROC 0.915 versus 0.922 for all 1,440 features. With the GC caliper, corresponding values were 0.864 and 0.882. The previously reported six-feature panel [9] attenuated to AUROC 0.661, compared with 0.950 in its unmatched comparison. These models distinguish the defined populations, but their performance depends strongly on sequence length. They do not classify peptide function.

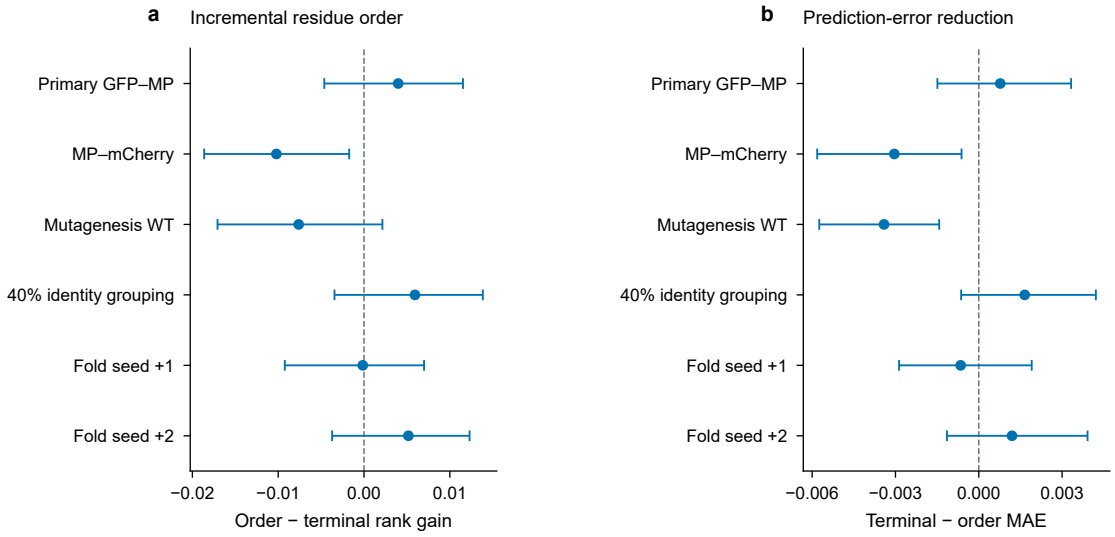
Broad clustering structure was computationally reproducible under the new fixed settings (S1 Fig). Family-weighted PCA at $k = 2$ had silhouette about 0.118, catalogue-label adjusted Rand agreement about 0.270, and native/shuffle agreement about 0.906. Seed agreement was perfect in that particular setting, with mean 80% subsampling agreement about 0.964. Finer clusters, family weights and original-space analyses were more sensitive. The available historical CSV and default feature code also do not reconcile with the reported retention count [9]. Thus exact historical membership reproduction remains unresolved, even though the controlled rerun succeeds.

4.2 Sequence edits and reporter context require separate transfer tests

A model that ranks unmodified peptides need not predict the effect of an edit or retain its performance in another reporter. We test paired edits, the original source population rules, identical peptides across reporter contexts and a separate expression assay.



S2 Fig. Peptide coverage and validation groups. a: available constructs versus the development set or strict transfer subset. The 1,072 development cargos are evaluated by grouped cross-validation, not declared externally independent. Additional and Yang strict subsets are separated from retrieved Guo groups as specified. b: development group sizes; the largest contains 156 cargos. Sequence/gene links can form large components, which are not broken to improve evaluation.



S3 Fig. Incremental residue-order prediction relative to the terminal-aware baseline. Positive values favor O: rank gain is O minus T, error reduction is T minus O. The 40% grouping and fold-seed rows use the primary GFP–MP outcomes. Bars are conditional 95% group-bootstrap intervals. Orientation and library changes are endpoint-context sensitivities. Intervals spanning zero do not constitute formal evidence of equivalence.

4.2.1 Calibration changes perturbation conclusions

Native ranking did not translate directly into intervention ranking. Raw PAP C30 differences correlated with measured Leu-to-Ala effects at $\rho = 0.320$ but with extension effects at -0.005. Training-WT-only calibration changed these to 0.374 and 0.124. The secondary full-sequence PAP transfer improved from raw correlations 0.381/0.057 to calibrated 0.450/0.322 for Leu-to-Ala and extension, respectively. Thus the exploratory raw extension result cannot support a blanket claim that the frozen model contains no useful extension information.

Calibrated full PAP achieved delta MAE 0.1296 and 0.1380, compared with out-of-group family-mean baselines 0.1485 and 0.1500 (Fig S4). WT-trained T achieved 0.1369/0.1668; adding order gave 0.1384/0.1691. For extension effects, learning that edit family explicitly improved T to 0.1398, whereas opposite-family training gave 0.1594. Opposite-family T correlations were -0.122 for Leu-to-Ala and 0.055 for extension. These results distinguish prediction of a native state, learning a perturbation family and transferring a correction between families.

AAC resemblance illustrates why score orientation matters: raw delta correlations were -0.133 for Leu-to-Ala and 0.202 for extension, but WT-fold calibration yielded 0.074 and -0.142. Its negative native association led the calibration to reverse direction. Prediction and observed change also share WT terms; baseline-state coupling and bounded reporter scales can affect delta correlations. We therefore retain same-library baselines, calibration directions and replicate-specific analyses, and make no new mechanistic claim from these associations. The mutation effects themselves were measured and reported by Guo and colleagues.

4.2.2 Original-source filtering qualifies the Leu-to-Ala contrast

On all 1,014 sequence-valid Leu-to-Ala pairs, WT-trained T exceeded A for mean-change ranking by $\Delta\rho = 0.097$ (95% conditional interval 0.054 to 0.148). Under the original final-two-Leu rule, A and T reached $\rho = 0.315$ and 0.301 on 798 pairs: their difference was -0.013 (-0.032 to 0.005; Table S1). Thus the wider-population terminal-feature advantage does not persist in this source-defined substitution population. Append-edit T-minus-A remained positive on both all 1,007 pairs (0.393, 0.321 to 0.470) and the 802 retained pairs (0.410, 0.323 to 0.495). Full-PAP source-WT calibration gains on the retained substitution and append populations were 0.093 (0.030 to 0.142) and 0.239 (0.186 to 0.301). O-minus-T intervals spanned zero in both. All 421 omitted family-specific pairs remain available in S12 Data. These are fixed-prediction, post-hoc population contrasts. They establish neither a causal effect of the exclusion nor a new terminal confound: the source authors already specified the rule [13].

Table S1. Original-source terminal-Leu sensitivity of mean-change prediction.

Edit	Population	Pairs	A	T
Leu-to-Ala	All valid	1,014	0.262	0.359
Leu-to-Ala	Source rule	798	0.315	0.301
Append	All valid	1,007	-0.246	0.147
Append	Source rule	802	-0.205	0.204

The source rule excludes WTs with Leu at either of the last two positions. All original fits and predictions are held fixed. A denotes composition; T adds terminal features. Values are Spearman correlations with same-library mutant-minus-WT mean scores. S12 Data retains both replicates, omitted strata, all original predictors and conditional group-bootstrap contrasts.

4.2.3 Identical peptides show reporter-context dependence

Guo and colleagues already compared GFP–MP and MP–mCherry reporter scores in their Supplementary Fig. S6A [13]. We extend that comparison to a common three-context cohort and evaluate fitted representation transfer while holding test peptides and dependence groups fixed. Matching the same 1,044 peptides removes sequence-cohort composition as the explanation for differences between the three Guo contexts. Observed G–M rank agreement was $\rho = 0.522$ (95% group-bootstrap interval 0.471 to 0.566), versus 0.867 (0.845 to 0.889) for G–U and 0.496 (0.440 to 0.540) for M–U (Fig 2). Within-context biological-replicate correlations were 0.963 for G ($n = 1,043$), 0.986 for M ($n = 1,041$), and 0.877 for U ($n = 1,039$). These replicate comparisons are descriptive and are not formal prediction ceilings or an estimate of a causal reporter effect.

Source-trained T retained positive cross-configuration ranking but performed below target-trained T on identical held-out groups (Table 2). G-trained T reached $\rho = 0.560$ on M, versus 0.742 for M-trained T; the paired rank gap was 0.182 (95% conditional interval 0.146 to 0.227). In the reverse direction, M-trained T reached 0.566 on G, versus 0.767 for G-trained T; the gap was 0.201 (0.163 to 0.244). Source-model calibration, learned from training groups, did not remove these target-scale error gaps: calibrated-transfer minus target-trained MAE was 0.0490 for G to M (0.0408 to 0.0572) and 0.0419 for M to G (0.0338 to 0.0492). Calibration is target-supervised adaptation, not an independent validation of a frozen predictor.

The composition-only model predicted across the original reporter configurations better than T: A reached $\rho = 0.663$ for G to M and 0.648 for M to G. Secondary paired contrasts of the saved predictions gave A-minus-T rank gains of 0.103 (95% conditional interval 0.078 to 0.132) and 0.082 (0.058 to 0.104), respectively. In contrast, T gave higher within-context point estimates than A for G (0.767 versus 0.746) and M (0.742 versus 0.728), although the corresponding incremental intervals included zero. Adding terminal features therefore did not consistently improve prediction across reporter contexts. These feature contrasts were examined after inspecting all A/T estimates and are explicitly secondary (S8 Data); they do not isolate a terminal degradation mechanism.

Prediction did not fail in every comparison: performance was better when the reporter stayed the same across libraries. G-trained T reached $\rho = 0.691$ on U, compared with 0.673 for U-trained T, with calibrated MAE 0.1211 versus 0.1305. In the reverse direction, U-trained T reached 0.737 on G, versus 0.767 for G-trained T. The stronger source-library replicate agreement is compatible with better source measurements helping prediction, but does not identify the cause of this asymmetry. The measured-source bridge required a source assay and achieved target MAE 0.0646 for G to U, compared with 0.2078 for G to M; it is reported separately from sequence models (S8 Data).

The T rank gap for G to M remained 0.161 with the 40% graph and 0.176 with the second split seed; the reverse gaps were 0.207 and 0.213. Both sensitivity settings also retained the G-to-U source-model advantage. The complete A-model, calibration, baseline and sensitivity results are retained in S8 Data. Reporter identity, fusion orientation, construct architecture and library conditions change jointly, so these observations cannot identify an isolated tag or terminal-exposure mechanism. They also do not explain the inverse Yang result or supply an independent replication.

4.2.4 Independent-study transfer fails to reproduce within-study ranking

The strict Yang subset contains 20 constructs, of which ten exceed the published 5% expression threshold. Primary MP–mCherry-trained T with FLAG sequence features had $\rho = -0.652$ (95% interval -0.888 to -0.246), and O had -0.650. Their AUROCs were 0.16 and 0.15, respectively. Direction was not inverted after observing validation outcomes. The negative result persists in the all-37 panel (T $\rho = -0.726$) and is retained as a failed transfer, rather than hidden by selecting another representation.

Native-only GFP–MP-trained T, a declared context sensitivity, had $\rho = 0.167$ (interval -0.321 to 0.616), whereas its FLAG-aware version had -0.303. FLAG-aware PAP C30 and full transfer had $\rho = -0.124$ and 0.083. Native-only full PAP had 0.359. The result therefore depends strongly on whether the input includes the tag and which reporter orientation supplied the training data (Fig S5). Neither representation gives convincing evidence of accurate prediction across these assays.

Supplied PepScore reached $\rho = 0.656$ (interval 0.308 to 0.865; AUROC 0.86), while native length alone reached 0.577 (interval 0.165 to 0.832; AUROC 0.83). The source describes random selection of the 29 uORFs across PepScores and adds eight high-scoring ORFs of other classes. The combined panel and validation originate in the PepScore publication. These estimates therefore do not provide an unbiased external comparison of PepScore with the other methods. A release-wide sequence audit matched ShortStop to 23 constructs ($\rho = -0.058$), including nine of the strict 20 ($\rho = 0.268$). This small covered subset cannot support a strong comparison with methods covering all 20; missing scores were not replaced by zero.

4.3 Independent transfer depends on the input and comparator

The independent cohorts test whether favorable within-study results extend to new sequences and interventions. Published predictors, matched sequence inputs and training-exposure audits determine which comparisons support a transfer claim.

4.3.1 Independent peptide transfer depends on source and decision metric

Composition models trained only on the source data transferred to the 14,143 eligible canonical tiles with $\rho = 0.816$ for GFP–MP training and 0.817 for MP–mCherry training (Table 3). The corresponding T correlations were 0.812 and 0.656. The GFP-trained A-minus-T difference was small and uncertain: 0.004 (95% conditional gene-bootstrap interval -0.001 to 0.010). The MP–mCherry gap was 0.161 (0.149 to 0.174). Thus this target supports a source-specific transfer penalty, not uniform harm from terminal features. At the stricter similarity threshold, the gaps were 0.007 (0.001 to 0.012) and 0.163 (0.151 to 0.176); the second source tuning seed retained the pattern (S9 Data).

Feature diagnostics qualify the interpretation (Fig 3). Removing PAP from T gave external $\rho = 0.793$ and 0.648, respectively. For GFP-trained heads, C4 gave 0.820, N4 0.778 and I4 0.793; for MP–mCherry training they gave 0.758, 0.685 and 0.765. Source grouped nested-CV correlations for A/T were 0.747/0.780 in GFP–MP and 0.728/0.728 in MP–mCherry. MP–mCherry N4 reached 0.763 in source CV but 0.685 externally. These diagnostic point estimates are compatible with context-specific positional information but do not separate feature complexity from exposure or other construct effects. Target-study-trained PAP internal/full references reached 0.824/0.862; their training provenance prevents a universal external-method ranking.

The secondary selection analysis gave a different GFP-trained comparison. T selected tiles with a higher mean measured assay score despite its slightly lower overall rank correlation: A-minus-T selected mean was −0.0145 (−0.0211 to −0.0073). For MP–mCherry-trained policies, A selected higher-scoring tiles: 0.0464 (0.0359 to 0.0564). Overall correlation alone therefore cannot show which model selects the peptides with the highest measured scores. These are retrospective policies on canonical peptide reporters, not evidence about native nORF abundance or biological function.

4.3.2 Strong published models reduce the apparent advantage of composition-based prediction

On the same 14,143 native tiles, frozen QCDPred reached $\rho = 0.739$ using complete-window means and 0.800 using normalized whole-cargo composition (Fig 4). The latter reduces the A-head advantage to about 0.016–0.017, with top-decile selected-mean difference intervals spanning zero (S10 Data). Same-study human25 and PAP references must be distinguished from these externally trained models. Published PAP CNN1 and CNN7 reached 0.873 and 0.877 on their author-held-out tiles. Thus useful nORF-to-tile composition transfer does not establish a new composition principle or superiority over the stronger target-study model.

4.3.3 Independent paired effects depend on representation and input coverage

Observed Chu replicate changes correlated at $\rho = 0.967$. On the primary 1,743-pair common cohort, GFP-trained A and T predicted mean paired changes at $\rho = 0.617$ and 0.547. The T-minus-A gap was −0.070 (95% conditional interval −0.103 to −0.039). The corresponding mCherry-source values were 0.622 and 0.343 (gap −0.278, interval −0.321 to −0.238). The poorer performance of these fitted models does not show that terminal information is always harmful: the frozen linear PAP C30 terminal component improved change ranking by 0.050 (0.030 to 0.068), whereas its full58 increment was 0.024 (−0.002 to 0.046).

Guo-WT-only calibration improved linear full58 PAP from $\rho = 0.651$ to 0.692, a gain of 0.041 (0.027 to 0.057; Table 4). The stronger published neural model narrows that finding: raw full58 CNN1 already reached 0.729, and calibrated CNN1 reached 0.735. Its calibration gain was only 0.006 (−0.002 to 0.014). CNN7 reached 0.725 raw and 0.725 calibrated, with gain −0.001 (−0.008 to 0.006; Fig S6). The post-hoc raw-CNN1-minus-calibrated-linear contrast was 0.037 (0.019 to 0.056). Calibration improved the linear model, but these results do not show that it generally improves the stronger neural models.

Input coverage had a larger effect than the added neural calibration in this cohort. Raw CNN1 reached 0.518 on C30 versus 0.729 on full58; the post-hoc gap was 0.211 (0.171 to 0.253). CNN7 reached 0.524 versus 0.725. This contrast does not isolate a biological cause, and full58 extrapolates beyond the training length. Both windows, all source contexts, both replicates, stricter similarity and transcript-match sensitivities remain reported. Human25 whole-cargo calibration did not improve its primary change correlation (0.662 raw versus 0.655 calibrated).

The conditional order test was positive for native held-out tiles: CNN7 native-minus-shuffle-mean $\Delta\rho = 0.0064$ (0.0050 to 0.0078). Independent paired-effect increments instead had intervals spanning zero: −0.0001 (−0.0110 to 0.0111) for C30 and 0.0057 (−0.0024 to 0.0129) for full58. The second shuffle seed gave the same pattern of results. Separately fitted full58 CNN7 and CNN1 did not establish a change-ranking advantage (post-hoc difference −0.0034, interval

−0.0180 to 0.0101). The model therefore captures a small residue-order signal on native tiles, but we found no clear improvement in predicting independent paired changes.

4.3.4 Terminal-model comparisons change with exposure and input coverage

The C23 development-screen audit matched 1,450 WT endpoints and six RT endpoints; their union covered 1,454 of the 1,743 common-cohort pairs. These pairs cannot be treated as equally independent terminal-model tests merely because they passed the separate source/PAP exclusions. Neither endpoint matched in 289 pairs/259 groups; the available-gene-symbol sensitivity retained 97 pairs/88 groups (Table S2).

Table S2. Terminal-model and PAP input-coverage comparisons. Entries are correlations with observed RT-minus-WT change on the same pairs within each row. C23 denotes the final 23 residues; full58 denotes the complete measured cargo. All-pair terminal results have development-screen exposure. The 97-pair sensitivity does not certify absence of gene or remote-homology overlap. All four strata, CNN7/C30 controls and component metrics are in S11 Data.

Exposure stratum	Pairs	C-model	CNN1 C23	CNN1 full58
All pairs	1,743	0.500	0.479	0.729
Neither endpoint matched	289	0.537	0.521	0.761
Also no available-symbol match	97	0.669	0.651	0.779

On all pairs, C-model change prediction reached $\rho = 0.500$, compared with 0.479 for CNN1 on identical C23 inputs and 0.729 for full58. The post-hoc same-C23 C-model-minus-CNN1 contrast was 0.021 (95% conditional interval −0.004 to 0.042). On the 289 endpoint-unmatched pairs, these values were 0.537, 0.521 and 0.761; the same-input contrast was 0.016 (−0.031 to 0.064; Fig 5). All matched-input contrast intervals span zero. Thus the large contrast against full58 cannot establish an architecture-only deficit of the terminal model. Equal input coverage still leaves different training sources and objectives.

C-model correlation rose to 0.669 in the 97-pair sensitivity, compared with 0.651 for CNN1 C23 and 0.779 for full58. The same-C23 gap was 0.018 (−0.067 to 0.117); the C-model-minus-full58 interval also spanned zero (−0.110, interval −0.269 to 0.048). Higher correlations after exclusion do not demonstrate exposure-caused inflation or absence of exposure effects. The populations differ, and the smaller sensitivity has wider uncertainty.

4.4 Exact candidate links define native follow-up experiments

The results identify candidates for experiments on individual peptides. We match existing reporter edits to full peptide sequences with published interaction and native-evidence annotations. These matches identify candidates to test; they do not show how the edits affect function.

4.4.1 Candidate interactions remain to be tested after editing

Of 60 selected Sandmann candidates, 24 had exact full-sequence matches in natural Guo records; 17 had primary GFP–MP measurements and 32 measured edit pairs (17 Leu-to-Ala, 15 append). Twenty-eight mean changes were positive and four negative; all were retained. This selected overlap is not a random sample or an estimate of functional enrichment. The GAS5 variant shares an identifier with four Guo records but differs in sequence, so none were joined. The Sandmann LINC01128 and PVT1 peptides had no exact match.

Three matched candidates carried five source domain/SLiM-associated WT binding tiles: RP11-620J15.3 (64 residues; CDK5-associated tile), JHDM1D-AS1 (68 residues; two CSNK2A1-associated tiles), and the SLCO5A1 uORF peptide (55 residues; AURKB- and SRPK2-associated tiles). Their Leu-to-Ala mean reporter changes were +0.578, +0.120 and +0.176, respectively; append changes were +0.126, +0.110 and +0.121. All six effects were positive in both source replicates, and all three WTs passed the source terminal-Leu rule. These effects are existing Guo measurements, newly linked here to the exact Sandmann peptide sequences. Global Leu-to-Ala changes residues within all five WT binding tiles; append retains their sequence. S12 Data provides every matched and unmatched candidate and residue-level annotation.

A source tagged-overexpression phosphorylation record places JHDM1D-AS1 Ser36 inside its kinase-associated tile region. This provides a separate observation about the candidate. It does not establish endogenous phosphorylation, identify the responsible kinase or show what happens after either edit. No edited binding, native turnover, peptide-dependent activity or stability–function tradeoff is established. The canonical SLCO5A1 transporter's activity is not assigned to its uORF. These candidates can be used to test whether an edit that increases reporter accumulation also preserves interaction. That experiment has not yet been performed.

4.4.2 Native-source support differs among the three candidates

Forty-nine of the 60 selected peptides matched the original heart tables exactly: 36 lncORFs and 13 uORFs. Fifteen had reported positive SRM records. The heart dataset helped identify candidates for the later panel; these counts describe coverage, not independent confirmation or functional enrichment. JHDM1D-AS1 had an in-vitro-translation flag, a shotgun peptide and three source-positive SRM runs across two hearts. The SRM signature `ASMEPPLVAGGLSPR` occupies residues 24–38. The runs had source dot products 0.7688, 0.7497 and 0.7045 under the original 0.7 threshold [34]. These source metrics do not supply a newly calibrated identification error rate or establish full-product ends or native turnover.

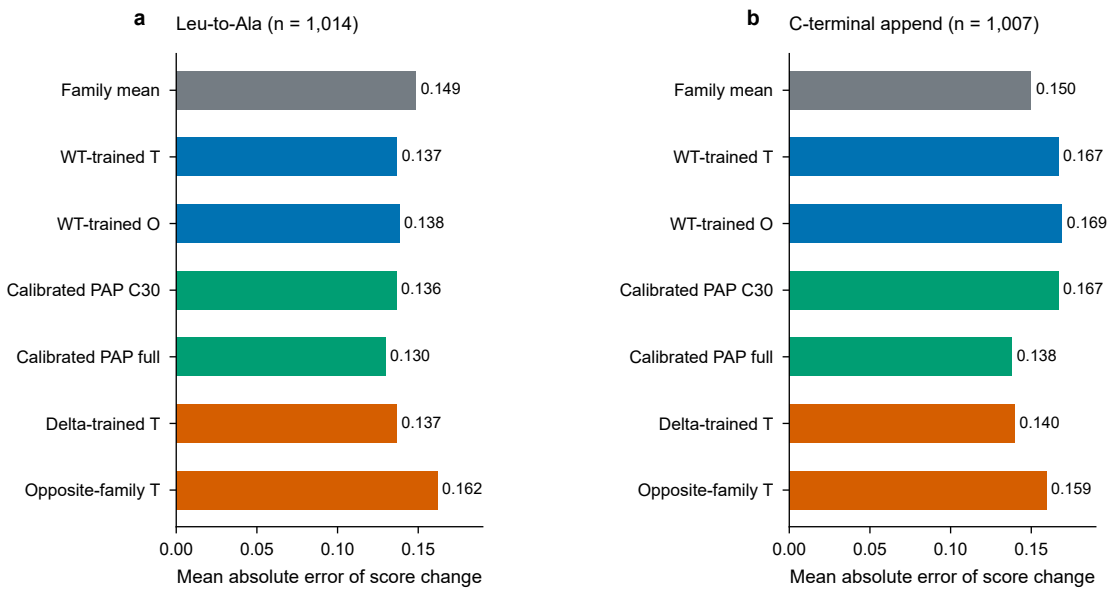
The later community reassessment materially qualifies the shotgun evidence. The exact JHDM1D-AS1 PSM for `CRGIQTLENVPPEFPASSK`, scan 21701 in PXD006675, has one recorded curator rating of 1/5, the scheme's poor-quality category [36]. This does not prove that the peptide is absent. The study explicitly excluded SRM from reassessment, so this rating cannot be transferred to the distinct SRM signature or its runs. The reported targeted evidence still requires claim-specific evaluation.

Forty-five of the 60 candidates matched the 2026 full-sequence catalogue. RP11-620J15.3, listed there as GIHCG/c12riboseqorf85, and the SLCO5A1 uORF/c8riboseqorf60 each have a source final Tier 2B entry with one HLA-I peptide [35]. RP11's `EVCLILKLK` has 10 reported PSMs but is also mapped to c12riboseqorf86; it cannot uniquely establish the selected 64-residue product. The uORF's `TPAPCGTRL` has one reported PSM. These single-HLA entries are not all manually validated by the catalogue's more selective annotation workflow. JHDM1D-AS1 has an exact full-sequence Table 9 row but no entry in the retrieved non-HLA, HLA or curated-annotation Tables 3, 7 or 12. Missing entries do not establish that translation or function is absent. HLA presentation does not by itself establish stable accumulation of a full cellular product.

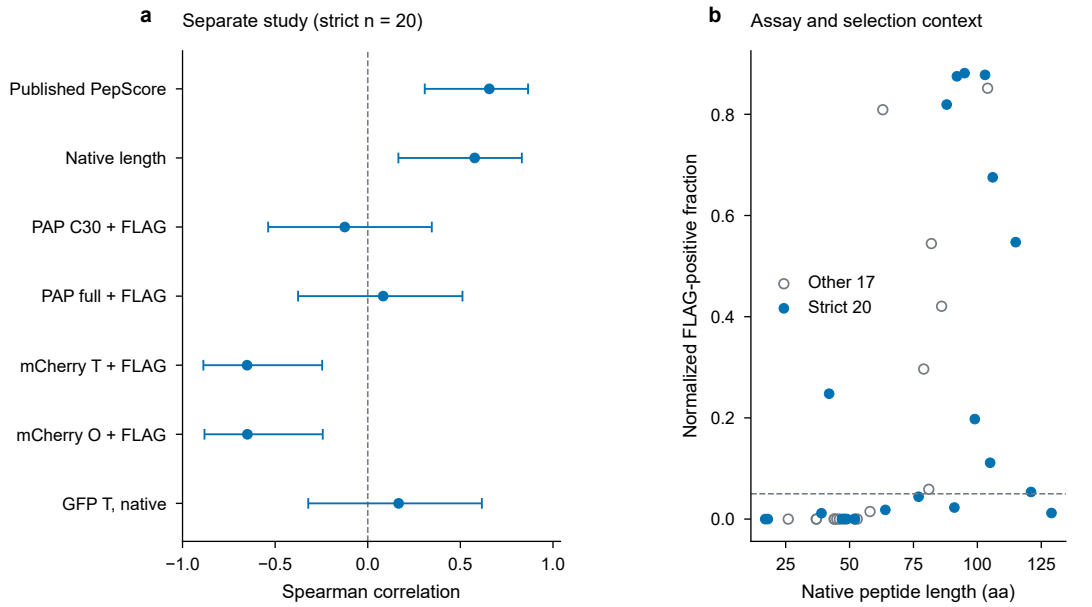
The selected JHDM1D-AS1 peptide is encoded exactly at nucleotides 338–541 of RefSeq NR_024451.1, with a TAA at 542–544 (1-based). This 2,398-nt reference differs in overall length from the 951-nt heart assembly. It is compatible with the reference cited by the 2017

RNA study [40], without verifying that study's plasmid or peptide expression. Its SRM signature overlaps both CSNK2A1-associated WT tiles and source tagged-overexpression Ser36. Leu-to-Ala changes the signature to `ASMEPPAVAGGASPR`, whereas append retains it. A WT-signature-only MS comparison would therefore confound an edited-product measurement with a changed analyte. Edit-specific standards or a separately validated common readout are needed before comparing product amounts.

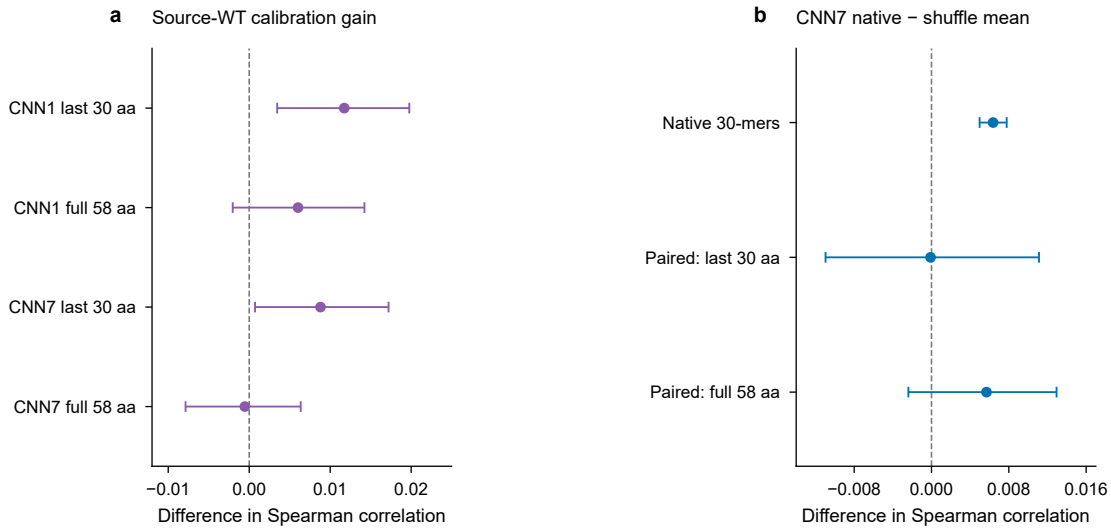
The brain-atlas comparison also distinguishes two peptides sharing the MKKS name. The selected 63-residue peptide exactly matches micro-MKKS63/Q9HB66, with a source strong DDA grade and 1,004 unique spectral counts; the separate 50-residue MKKS peptide is listed without MS evidence [38]. All 60 candidates were retained, with nine exact Table S2 matches overall. RP11-620J15.3 is present in the atlas's RiboCode/ShortStop-selected subset without MS evidence, rather than as an independently confirmed protein. Neither MKKS peptide has an exact Guo assay match. Micro-MKKS63 could therefore serve as a known native-product control in new experiments. It provides neither an existing WT/edit comparison nor a new functional finding (S15 Data).



S4 Fig. Prediction of within-parent reporter-score changes. All WT-trained models and isotonic calibrations exclude the evaluated parent group. The family-mean baseline also excludes that group. Delta-trained models fit changes from other parents in the same family; opposite-family models use the other edit family, again excluding test-parent groups. PAP full is a declared secondary sequence transfer. Raw incompatible score-scale MAE is not plotted. Ranking, replicate sensitivities and uncertainty are supplied separately. This figure retains all sequence-valid pairs; Table S1 and S12 Data add the original-source terminal-Leu restriction.



S5 Fig. Transfer to a separate FLAG-expression assay. a: strict $n = 20$ rank correlations with conditional group-bootstrap intervals. The primary direct transfer uses MP–mCherry training and actual FLAG-fusion features. Native-only GFP–MP transfer is a declared sensitivity, not a post-validation replacement. PepScore and length refer to the source's mixed panel; this is not an unbiased method leaderboard. b: all 37 outcomes and native lengths, highlighting the strict subset. The dashed line is the authors' 5% expression threshold. Absolute errors are not compared across assay scales.



S6 Fig. Calibration and conditional-order increments depend on the target. (a) Calibrated-minus-raw paired-effect correlation for each neural model/input. Both full58 intervals span zero. (b) CNN7 native-minus-shuffle-mean rank contrasts preserve composition and the first/final five residues. The native 30-mer result is an author-held-out target-study reference; paired-effect results use the independent Chu common cohort. Bars are 95% conditional group-bootstrap intervals from 1,000 resamples, without multiplicity adjustment. The second shuffle seed and all raw/calibrated representations are in S10 Data. These are predictive controls, not measured biological shuffle effects.

References

[1] Mudge JM et al. Standardized annotation of translated open reading frames. Nature Biotechnology. 2022. doi:10.1038/s41587-022-01369-0.

[2] Chothani S et al. An expanded reference catalog of translated open reading frames for biomedical research. Nucleic Acids Research. 2026;54:gkag234. doi:10.1093/nar/gkag234.

[3] Samandi S et al. Deep transcriptome annotation enables the discovery and functional characterization of cryptic small proteins. eLife. 2017;6:e27860. doi:10.7554/eLife.27860.

[4] Brown A, Rathore S, Kimanius D, Aibara S, Bai XC, Rorbach J, Amunts A, Ramakrishnan V. Structures of the human mitochondrial ribosome in native states of assembly. Nature Structural and Molecular Biology. 2017;24:866–869. doi:10.1038/nsmb.3464.

[5] Rathore A et al. MIEF1 Microprotein Regulates Mitochondrial Translation. Biochemistry. 2018;57:5564–5575. doi:10.1021/acs.biochem.8b00726.

[6] Shepelev NM et al. Translation of numerous upstream open reading frames rather than their products is essential for proliferation of human cells of distinct origin. bioRxiv [Preprint]. 2025 July 7;version 1. doi:10.1101/2025.07.07.663426.

[7] Ghosh A et al. Suppressive cancer nonstop extension mutations increase C-terminal hydrophobicity and disrupt evolutionarily conserved amino acid patterns. Nature Communications. 2024;15:9209. doi:10.1038/s41467-024-52779-4.

[8] Hasenjäger S, Bologna A, Essen LO, Spadaccini R, Taxis C. C-terminal sequence stability profiling in Saccharomyces cerevisiae reveals protective protein quality control pathways. Journal of Biological Chemistry. 2023;299:105166. doi:10.1016/j.jbc.2023.105166.

[9] Gounder R. The Elusive Dynamic of the Genome: Novel Sequence Patterns, Splicing and Regulation. PhD thesis, University of Cambridge. 2025. Chapter 2 and Appendix A; supplied thesis PDF.

[10] Johansson KE, Mashahreh B, Hartmann-Petersen R, Ravid T, Lindorff-Larsen K. Prediction of quality-control degradation signals in yeast proteins. Journal of Molecular Biology. 2023;435:167915. doi:10.1016/j.jmb.2022.167915.

[11] Mashahreh B et al. Conserved degronome features governing quality control associated proteolysis. Nature Communications. 2022;13:7588. doi:10.1038/s41467-022-35298-y.

[12] Zhang Z et al. Elucidation of E3 ubiquitin ligase specificity through proteome-wide internal degron mapping. Molecular Cell. 2023;83:3377–3392.e6. doi:10.1016/j.molcel.2023.08.022.

[13] Guo Y et al. Intrinsic bias of the genetic code shapes the folding and stability landscapes of microproteins. Molecular Cell. 2026;86:2173–2190.e11. doi:10.1016/j.molcel.2026.04.021. Data: doi:10.17632/jjxy3f462r.1.

[14] Voutsinos V et al. A near-complete map of human cytosolic degrons and their relevance for disease. Science Advances. 2026;12:eadz3483. doi:10.1126/sciadv.adz3483.

[15] Chu CY et al. Protein C-terminal variations impact proteostasis. Nature Communications. 2026;17:2288. doi:10.1038/s41467-026-68979-z.

[16] Casola C, Owoyemi A, Vakirlis N. Degradation determinants are abundant in human non-canonical proteins and minor annotated isoforms. Genome Biology and Evolution. 2026. doi:10.1093/gbe/evag009.

[17] Roginski P, Papadopoulos C, Herman S, Baumann A, Grislain A, Lopes A. Impact of GC content on de novo gene birth. Nature Communications. 2026;17:1268. doi:10.1038/s41467-025-68022-7.

[18] Szulc NA et al. DEGRONOPEDIA: a web server for proteome-wide inspection of degrons. Nucleic Acids Research. 2024. doi:10.1093/nar/gkae238.

[19] Koren I et al. The eukaryotic proteome is shaped by E3 ubiquitin ligases targeting C-terminal degrons. Cell. 2018;173:1622–1635.e14. doi:10.1016/j.cell.2018.04.028.

[20] Timms RT et al. A glycine-specific N-degron pathway mediates the quality control of protein N-myristoylation. Science. 2019;365:eaaw4912. doi:10.1126/science.aaw4912.

[21] Schulze TK et al. Supervised learning of protein variant effects across large-scale mutagenesis datasets. Protein Science. 2026. doi:10.1002/pro.70526.

[22] Tareen A, Kooshkbaghi M, Posfai A, Ireland WT, McCandlish DM, Kinney JB. MAVE-NN: learning genotype-phenotype maps from multiplex assays of variant effect. Genome Biology. 2022;23:98. doi:10.1186/s13059-022-02661-7.

[23] Faure AJ, Lehner B. MoCHI: neural networks to fit interpretable models and quantify energies, energetic couplings, epistasis, and allostery from deep mutational scanning data. Genome Biology. 2024;25:303. doi:10.1186/s13059-024-03444-y.

[24] Faure AJ, Domingo J, Schmiedel JM, Hidalgo-Carcedo C, Diss G, Lehner B. Mapping the energetic and allosteric landscapes of protein binding domains. Nature. 2022;604:175–183. doi:10.1038/s41586-022-04586-4.

[25] LeBlanc CJ et al. Interpretable biophysical neural networks of transcriptional activation domains separate roles of protein abundance and coactivator binding. Cell Systems. 2026;17:101701. doi:10.1016/j.cels.2026.101701.

[26] Johansson KE, Lindorff-Larsen K, Winther JR. Global analysis of multi-mutants to improve protein function. Journal of Molecular Biology. 2023;435:168034. doi:10.1016/j.jmb.2023.168034.

[27] Miller B et al. ShortStop: a machine learning framework for microprotein discovery. BMC Methods. 2025;2:16. doi:10.1186/s44330-025-00037-4. Precomputed release v1.2.0, 28 October 2026: `https://github.com/brendan-miller-salk/ShortStop/releases/tag/v1.2.0`.

[28] Yang H, Li Q, Stroup EK, Wang S, Ji Z. Widespread stable noncanonical peptides identified by integrated analyses of ribosome profiling and ORF features. Nature Communications. 2024;15:1932. doi:10.1038/s41467-024-46240-9.

[29] He J, Sui J, Chen Q, Hu H, Tan CSH. sORF-Trans2MS: a two-module deep learning framework for sORF translation and MS-supported microprotein prediction. bioRxiv. 2026. Version 1; preprint. doi:10.64898/2026.09.29.755517.

[30] Fierro-Monti I. Translation uniformity, evolutionary conservation, and peptide properties associate with microprotein mass spectrometry detection. Research Square. 2026. Version 1; preprint. doi:10.21203/rs.3.rs-9645389/v1.

[31] Xiao N, Cao DS, Zhu MF, Xu QS. protr/ProtrWeb: R package and web server for generating various numerical representation schemes of protein sequences. Bioinformatics. 2015;31:1857–1859. doi:10.1093/bioinformatics/btv042.

[32] Hickey CM, Breckel C, Zhang M, Theune WC, Hochstrasser M. Protein quality control degron-containing substrates are differentially targeted in the cytoplasm and nucleus by ubiquitin ligases. Genetics. 2021;217:iyaa031. doi:10.1093/genetics/iyaa031. Supplementary Table 5: doi:10.25386/genetics.13273286.

[33] Sandmann CL et al. Evolutionary origins and interactomes of human, young microproteins and small peptides translated from short open reading frames. Molecular Cell. 2023;83:994–1011.e18. doi:10.1016/j.molcel.2023.01.023.

[34] van Heesch S et al. The translational landscape of the human heart. Cell. 2019;178:242–260.e29. doi:10.1016/j.cell.2019.05.010.

[35] Deutsch EW et al. Expanding the human proteome with microproteins and peptideins. Nature. 2026;654:813–825. doi:10.1038/s41586-026-10459-x.

[36] Wacholder A et al. Community benchmarking and evaluation of human unannotated microprotein detection by mass spectrometry based proteomics. Nature Communications. 2026;17:1241. doi:10.1038/s41467-025-68002-x.

[37] Wacholder A et al. Author Correction: Community benchmarking and evaluation of human unannotated microprotein detection by mass spectrometry based proteomics. Nature Communications. 2026;17:4882. doi:10.1038/s41467-026-73431-3.

[38] Miller B et al. A microprotein atlas of the human frontal cortex in Alzheimer's disease. Nature Aging. 2026;6:1979–1997. doi:10.1038/s43587-026-01207-x. Source repository: `https://github.com/brendan-miller-salk/brain-microprotein-atlas`.

[39] Schlesinger D et al. A large-scale sORF screen identifies putative microproteins involved in cancer cell fitness. iScience. 2025;28:111884. doi:10.1016/j.isci.2025.111884.

[40] Kondo A et al. Long noncoding RNA JHDM1D-AS1 promotes tumor growth by regulating angiogenesis in response to nutrient starvation. Molecular and Cellular Biology. 2017;37:e00125-17. doi:10.1128/MCB.00125-17.

[41] Larsen FB et al. Comprehensive degron mapping in human transcription factors. Science Advances. 2026;12(45):eaee1766. doi:10.1126/sciadv.aee1766.

[42] Salinas VH, Ranganathan R. Coevolution-based inference of amino acid interactions underlying protein function. eLife. 2018;7:e34300. doi:10.7554/eLife.34300.

[43] Faure AJ, Martí-Aranda A, Hidalgo-Carcedo C, Beltran A, Schmiedel JM, Lehner B. The genetic architecture of protein stability. Nature. 2024;634:995–1003. doi:10.1038/s41586-024-07966-0.

[44] Martí-Aranda A, Lehner B. Seven complete comparative maps of allosteric mutations in a protein family. Nature Communications. 2026;17:5487. doi:10.1038/s41467-026-71005-x.

[45] Hidalgo-Carcedo C, Faure AJ, Martí-Aranda A, Zarin T, Lehner B. Allosteric and energetic remodeling of a PDZ domain by protein domain extensions. Nature Communications. 2026;17:2934. doi:10.1038/s41467-026-69673-w.

[46] Fawzy M, Marsh JA. Understanding the heterogeneous performance of variant effect predictors across human protein-coding genes. Scientific Reports. 2024;14:26114. doi:10.1038/s41598-024-76202-6.

[47] Guharoy M et al. Tripartite degrons confer diversity and specificity on regulated protein degradation in the ubiquitin-proteasome system. Nature Communications. 2016;7:10239. doi:10.1038/ncomms10239.

[48] Guharoy M, Lazar T, Macossay-Castillo M, Tompa P. Degron masking outlines degronons, co-degrading functional modules in the proteome. Communications Biology. 2022;5:445. doi:10.1038/s42003-022-03391-z.

[49] Middendorf L, Ravi Iyengar B, Eicholt LA. Sequence, Structure, and Functional Space of Drosophila De Novo Proteins. Genome Biology and Evolution. 2024;16(8):evae176. doi:10.1093/gbe/evae176.

[50] Ruiz-Orera J et al. Evolution of translational control and the emergence of genes and open reading frames in human and non-human primate hearts. Nature Cardiovascular Research. 2024;3:1217. doi:10.1038/s44161-024-00544-7.

[51] Vakirlis N, Vance Z, Duggan KM, McLysaght A. De novo birth of functional microproteins in the human lineage. Cell Reports. 2022;41:111808. doi:10.1016/j.celrep.2022.111808.

[52] Chen J et al. Pervasive functional translation of non-canonical human open reading frames. Science. 2020;367:1140–1146. doi:10.1126/science.aay0262.

[53] Prensner JR et al. Non-canonical open reading frames encode functional proteins essential for cancer cell survival. Nature Biotechnology. 2021;39:697–704. doi:10.1038/s41587-020-00806-2.

[54] Hofman DA et al. Translation of non-canonical open reading frames as a cancer cell survival mechanism in childhood medulloblastoma. Molecular Cell. 2024;84:261–276.e18. doi:10.1016/j.molcel.2023.12.003.

[55] Salghetti SE, Muratani M, Wijnen H, Futcher B, Tansey WP. Functional overlap of sequences that activate transcription and signal ubiquitin-mediated proteolysis. Proceedings of the National Academy of Sciences USA. 2000;97:3118–3123. doi:10.1073/pnas.97.7.3118.

[56] Matreyek KA et al. Multiplex assessment of protein variant abundance by massively parallel sequencing. Nature Genetics. 2018;50:874–882. doi:10.1038/s41588-018-0122-z.

[57] Yao G et al. Long non-coding RNA JHDM1D-AS1 interacts with DHX15 protein to enhance non-small-cell lung cancer growth and metastasis. Molecular Therapy Nucleic Acids. 2019;18:831–840. doi:10.1016/j.omtn.2019.09.028.

[58] Malmström E et al. The long non-coding antisense RNA JHDM1D-AS1 regulates inflammatory responses in human monocytes. Frontiers in Cellular and Infection Microbiology. 2022;12:934313. doi:10.3389/fcimb.2022.934313.

[59] Liu C et al. LncRNA RP11-620J15.3 promotes HCC cell proliferation and metastasis by targeting miR-326/GPI to enhance glycolysis. Biology Direct. 2023;18:15. doi:10.1186/s13062-023-00370-0.

[60] Devaux A et al. Identification of microprotein-coding intronic polyadenylation isoforms and function in genotoxic anticancer drug response. Genome Biology. 2025;26:366. doi:10.1186/s13059-025-03829-7.